الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني

الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام ٢٠٢٣ الثانية

الاسم والشهرة: ----------

رقم المرشح: ----------

الاختصاص والرمز : الالكترونيك (٢٠٢٠٣)
المادة : الالكترونيك الرقمي
المستندات المسموح بها : آلة حاسبة غير مبرمجة

المعدل : (١٨)
المدة : ساعتان

الإجابات على "كراس الإجابة"

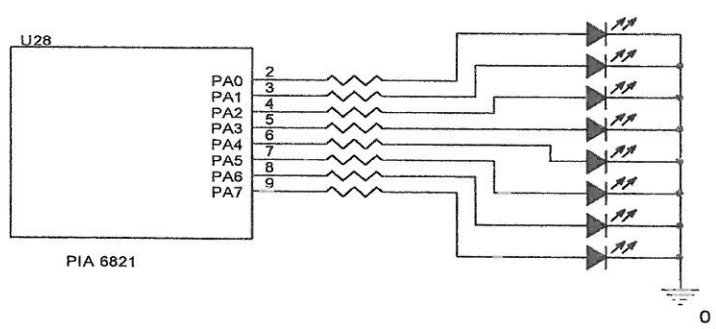
A- Répondre par Vrai ou Faux: (½pt/ réponse correcte)

1- Si le registre DDRA = (1111 1111)2 alors le PORTA est initialisé à 4 entrées et 4 soties.
2- Si le registre DDRA = $(0000\ 1111)_B$ alors le PORTA est totalement initialisé en entrées.
3- Le résultat de l'instruction [ANDB #$FF] avec (B) = $(FF)_{16}$ est: (B) = $(FF)_{16}$.
4- Le résultat de l'instruction [ADDB #$FF] avec (B) = $(FF)_{16}$ est (B) = $(FF0)_{16}$.
5- Pour obtenir la différence $(180 - 70)_{10}$ avec (B) = $(180)_{10}$, on écrit l'instruction suivante: SUBB #$250.
6- Pour allumer LED1 et LED2 connectées aux broches PB1 et PB2 du PORTB par leurs anodes, on écrit LDB #$06 et STB ORB.
7- Le résultat de l'instruction [EORA #$AA] avec (A) = (FF) est : (A) = $(55)_{16}$
8- La mémoire $0040 à la fin de ce programme : [(LDB#$02) ; (LDA#$05) ; (MUL) ; (STB $0040) ; END] contient $(0A)_{16}$.
9- Le résultat de l'instruction [ORA #$AA] avec (A) = $FF est: (A) = $00
10- Le résultat de l'instruction [LDA $40] avec ($40) = $(1F)_H$ est: (A) = $40

B- Traiter 2 questions seulement: (7½ pts / question)

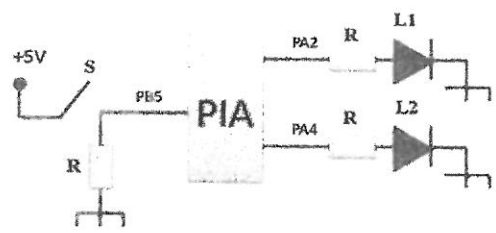
I- Ecrire le programme qui affiche séquentiellement sur le port B d'un PIA6821 connecté au MP6809, les octets du tableau suivant :(voir figure)

PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
1	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	0	0	1	0	0	0
0	0	0	0	0	1	0	0
0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	1



II- Ecrire un programme en langage assembleur 6809 pour exécuter ce qui suit:

1- S est ouvert ➔ L1 allumée et L2 éteinte
2- S est fermé ➔ L1 éteinte et L2 allumée



III- 1- Considérer le registre suivant:

a- Quel est son nom ?
b- Expliquer le rôle de chacun de ses bits (0 à 7).

7	6	5	4	3	2	1	0
			I	N	Z	V	C

2- Expliquer le rôle de chacun des registres suivants:

a- Accumulateur
b- Registre d'index
c- Pointeur de pile
d- Compteur de programme (compteur ordinal)

3- Commenter l'opération de chaque instruction et indiquer le résultat de chacune de séquences suivantes :

a-
```
LDA #$24
SUBA #$04
ASRA
```

b-
```
LDB #$2F
INCB
```

ANNEXE / APPENDIX

Instruction set of 6809

ADCA, ADCB, ADDA, ADDB, ADDD, ANDA, ANDB, ANDCC, ASLA, ASLB, ASL, ASRA, ASRB, ASR, BITA, BITB, CLRA, CLRB, CLR, CMPA, CMPB, CMPD, CMPU, CMPS, CMPX, CMPY, COMA, COMB, COM, CWAI, DAA, DECA, DECB, DEC, EORA, EORB, EXG, INCA, INCB, INC, JMP, JSR, LDA, LDB, LDD, LDS, LDU, LDX, LDY, LEAS, LEAU, LEAX, LEAY, LSLA, LSLB, LSL, LSRA, LSRB, LSR, MUL, NEGA, NEGB, NEG, NOP, ORA, ORB, ORCC, PSHS, PSHU, PULS, PULU, ROLA, ROLB, ROL, RORA, RORB, ROR, RTI, RTS, SBCA, SBCB, SEX, STA, STB, STD, STS, STU, STX, STY, SUBA, SUBB, SUBD, SWI, SWI2, SWI3, SYNC, TFR, TSTA, TSTB, TST.

PIA:

ORA et DDRA = $8000	**CRA = $8001**
ORB et DDRB = $8002	**CRB = $8003**

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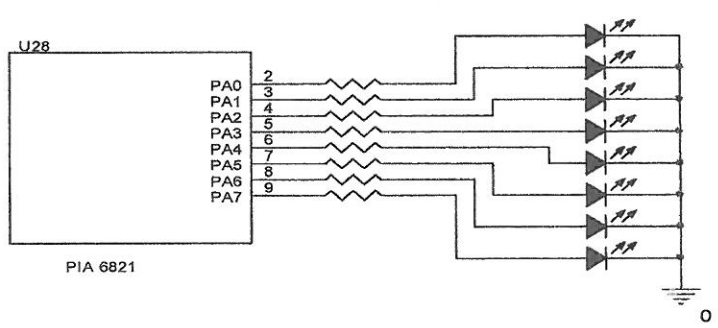
A- Answer by true or false: (½pt / correct answer)

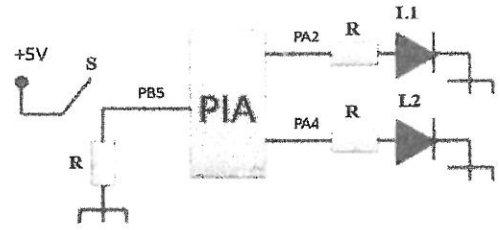
1- If the register DDRA = (11111 1111)$_2$ then PORTA is initialized as 4 inputs and 4 outputs.
2- If the register DDRA = (0000 1111)$_B$ then PORTA is totally initialized as inputs.
3- The result of the instruction [ANDB #$FF] with (B) = (FF)$_{16}$ is: (B) = (FF)$_{16}$.
4- The result of the instruction [ADDB #$FF] with (B = (FF)$_{16}$ is: (B) = (FF0)$_{16}$.
5- To get the difference (180 – 70)$_{10}$ with (B) = (180)$_{10}$, we write the following instruction: SUBB #$250.
6- To turn ON LED1 and LED2 connected to pins PB1 and PB2 of PORTB by their anodes, we write LDB #$06 and STB ORB.
7- The result of the instruction [EORA #$AA] with (A) = (FF)$_{16}$ is : (A) = (55)$_{16}$
8- The content of the memory $0040 at the end of this program:[(LDB#$02) ; (LDA#$05) ; (MUL) ; (STB $0040) ; END] is (0A)$_{16}$
9- The result of the instruction [ORA #$AA] with (A) = $FF is : (A) = $00
10- The result of the instruction [LDA $40] with ($40) = (1F)$_H$ is : (A)=$40

B- Treat 2 questions only: (7½ pts / question)

I- Write the program that displays sequentially at port B of a PIA6821 connected to the MP6809, the bytes of the following table: (see figure)

PB7	PB6	PB5	PB4	PB3	PB2	PB1	PB0
1	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	0	0	1	0	0	0
0	0	0	0	0	1	0	0
0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	1

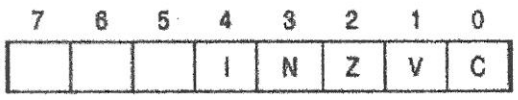




II- Write a program in 6809 assembly language to execute the following:

1- S is open ➔ L1 ON and L2 OFF
2- S is closed ➔ L1 OFF and L2 ON

III- 1- Consider the following register:



a- What is its name?
b- Explain the role of each one of its bits (0 to 7).

2- Explain the role of each one of the following registers:

a- Accumulator
b- Index register
c- Stack pointer
d- Program counter

3- Comment the operation of each instruction and indicate the result of each of the following sequences:

a-
```
LDA #$24
SUBA #$04
ASRA
```

b-
```
LDB #$2F
INCB
```

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PIA:

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